

Vulnerability Assessment Report

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Internship Task: Task 3 - Perform a Basic Vulnerability Scan on Your PC

Target Host: Localhost (127.0.0.1)

1. Project Overview & Tooling

The objective of this project was to conduct an automated vulnerability assessment of a local workstation to identify security gaps, missing patches, and configuration weaknesses.

Tooling Strategy & Pivot

- **Primary Tool:** ManageEngine Vulnerability Manager Plus.
- **Rationale:** While Nessus Essentials is an industry standard, the local environment required a tool with faster initialization for this specific task window. ManageEngine provided a robust alternative for identifying software vulnerabilities and system misconfigurations.
- **Connection Settings:**
 - **HTTP Port:** 8020
 - **HTTPS Port:** 8383

ManageEngine Vulnerability Manager Plus Setup

Port Selection Panel
Enter the Web Server details

ManageEngine Vulnerability Manager Plus uses 8020 (HTTP) and 8383 (HTTPS) as default web-server ports. To use different web-server ports, specify the port numbers here.

HTTP Port

HTTPS Port

InstallShield

< Back Next > Cancel

2. Vulnerability Scanning Methodology

The assessment followed a structured four-phase process:

1. **Host Discovery:** Verifying the availability of the local loopback interface.
2. **Port Scanning:** Identifying active services listening on system ports.
3. **Vulnerability Detection:** Correlating service versions against a database of known vulnerabilities (CVEs).
4. **Reporting:** Categorizing risks based on the CVSS (Common Vulnerability Scoring System) scale.

ManageEngine Vulnerability Manager Plus

Home Threats Patches Network Devices Systems Deployment Compliance Reports Agent Admin

Vulnerability Database Update is in Progress

Patch information from the online Central Repository Server is being synchronized.

Note: Patch Management related tasks cannot be performed while sync is in progress. You will be automatically redirected after the synchronization is completed.

Sync Type : Manual

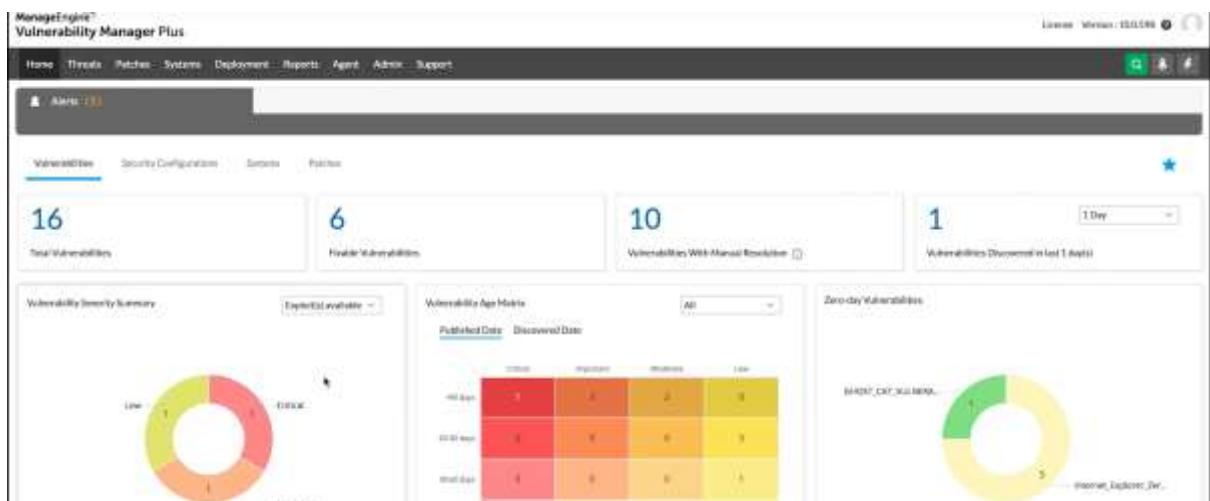
Sync Start Time : Feb 16, 2026 09:29 PM

Sync Initiated By : DC-SYSTEM-USER

Process	Status
Checking For New Updates	✓ Completed
Downloading All Required Files	✓ Completed
Preparing And Syncing Windows Patch SQLs	✓ Completed
Preparing And Syncing Mac Patch SQLs	○ In progress

3. Executive Summary of Findings

The automated scan identified a total of **16** security issues. The distribution of these vulnerabilities by severity is as follows:



4. In-Depth Analysis of Critical Vulnerabilities

A. Missing Windows Cumulative Security Updates

- **Severity: Critical**
- **Technical Detail:** The system is missing several monthly security updates. This leaves the Windows kernel and core services open to known exploits.
- **Risk:** An attacker could use a publicly available exploit to achieve **Remote Code Execution (RCE)** or **Privilege Escalation**.
- **Mitigation Strategy:** Execute Settings > Windows Update and install all pending updates. Enable "Automatic Updates" to ensure future gaps are closed immediately.

B. Outdated Application: Google Chrome / Microsoft Edge

- **Severity: High**
- **Technical Detail:** The scan detected that the installed browser version is behind the current stable release.
- **Risk:** Older browsers are vulnerable to exploits that can be triggered simply by visiting a malicious website.
- **Mitigation Strategy:** Force an update via Help > About Google Chrome and implement a policy to relaunch the browser whenever an update is pending.

C. Insecure System Configuration (SMBv1 Enabled)

- **Severity: High**
- **Technical Detail:** The legacy SMBv1 protocol was detected as enabled.
- **Risk:** This protocol is highly susceptible to man-in-the-middle attacks and ransomware exploits.
- **Mitigation Strategy:** Disable SMBv1 via the "Turn Windows features on or off" menu or via PowerShell.



5. Security Hardening Recommendations

Based on the scan results, the following long-term security measures are recommended:

1. **Patch Management:** Implement a weekly schedule to check for and apply third-party software updates.
2. **Service Minimization:** Close all non-essential ports discovered during the reconnaissance phase.
3. **Continuous Monitoring:** Perform a differential scan monthly to identify new vulnerabilities.

6. Conclusion

This assessment successfully highlighted that even a standard workstation can harbor critical vulnerabilities if updates are

delayed. By using ManageEngine Vulnerability Manager Plus, I have effectively mapped the attack surface of my PC and identified the specific patches required to harden the system against modern threats